Enabling Trust in Automotive IoT: Lightweight Mutual Authentication Scheme for Electronic Connected Devices in Internet of Things

Muhammad Nawaz Khan, Haseeb Ur Rahman, Tariq Hussain, Bailin Yang, and Saeed Mian Qaisar

***Abstract*—The emergence of smart and embedded devices and the adaptation of new technologies with the Internet of Things have entirely changed online shopping with a new e-commerce paradigm. However, while the accessibility of IoT services has many advantages, it also creates hazards to customer-related data. Due to pervasive e-commerce services, anyone can intercept and be compromised, creating great security and privacy concerns. To address these security challenges and to provide lightweight authentication for all entities, we introduce a "Lightweight Mutual Authentication (LMA) Scheme for Connected Devices in the IoT". The proposed scheme uses automatic and mutual authentication for all entities, employing a distributed approach within a server-based architecture. It is lightweight because it provides a secure way of using e-services with fewer steps, and it is automotive because the entities automatically authenticate each other. The LMA is formally validated in CTL, and the experimental results show that it decreases computation cost by about 56%, increases throughput by about 33.3%, and communication cost remains the same as the average of the other three schemes. In evaluation, the results demonstrate that the presence of LMA leads to a 20-millisecond increase in delay and a 2% decrease in PDR for 100 devices.**

***Index Terms*—Authentication, Internet of Things, security mechanisms, electronics devices, lightweights.**

I. Introduction

THE INTERNET of Things (IoT) is a sophisticated network encompassing interconnected objects, services, humans, and devices that engage in communication [1], [2], [3], and data exchange to achieve common objectives across diverse domains and applications [4], [5], [6]. This integrated network has had a transformative effect on several sectors, including but not limited to transportation, agriculture, healthcare, and energy generation and transmission. The primary objective of the IoT is to revolutionize our daily lives by enabling intelligent devices to perform tasks. Concepts like smart homes, smart cities, and smart infrastructure are closely associated with the IoT, as they leverage its capabilities to enhance efficiency and convenience [7], [8], [9], [10]. The domains of IoT expand across personal, social, corporate, and utility monitoring contexts in the field of linked electronic devices. The revolutionary potential of IoT is found in its dedication to offering cutting-edge services to society and enterprises, drastically transforming our interactions with the world around us. This broad goal entails integrating embedded electronics and information technologies into items worldwide. This integration enables these items to create significant information and serve as vital components in a network [11], [12], [13], [14]. By leveraging technologies such as embedded systems, processors, and communication systems, even the smallest elements of our environment can be interconnected based on specific applications and business needs. The result is a vast inter-networking environment that surpasses the scale of the current network infrastructure, offering unparalleled possibilities for connectivity, data exchange, and innovation, particularly within the context of consumer electronic connected devices [15]. With the expansion of the IoT, we can expect a future where every object becomes a potential source of information and contributes to the interconnected fabric of our digitally driven world [10].

In the personal and social domain, IoT applications allow users to interact with their surroundings and facilitate the maintenance and development of social relationships [16]. In the realm of connected consumer electronic devices, various sectors harness the capabilities of IoT for diverse applications. The transportation domain utilizes smart cars, roads, and traffic signals to ensure safe and convenient transportation facilities. Enterprises and industries employ IoT applications in finance, banking, marketing, and other areas to enable efficient inter- and intra-organizational activities [17], [18], [19]. Lastly, the service and utility monitoring sector leverages IoT for applications in agriculture, breeding, energy management, recycling operations, and more. The diverse application domains of IoT demonstrate its immense potential to transform various aspects

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of our lives. By integrating intelligent technologies and connectivity, the IoT enables enhanced productivity, sustainability, and resource management across multiple sectors. As IoT continues to evolve, it will unlock further opportunities for innovation and create a more interconnected and intelligent world [20], [21], [22].

In e-commerce-connected electronic devices, security considerations are important despite the various applications and advantages of connected devices [23], [24], [25]. While IoT offers various services across various fields and enhances human comfort, its inherent and dynamic nature raises the importance of authentication for every entity involved. The decentralized and interconnected nature of the IoT presents vulnerabilities that malicious intruders can exploit [26]. Without proper authentication mechanisms, unauthorized entities may gain access to sensitive information, manipulate devices, or disrupt critical systems. Therefore, ensuring the authenticity and trustworthiness of each entity becomes crucial in maintaining the integrity and security of the IoT infrastructures, especially in consumer e-commerce [10], [27], [28]. Authentication serves as a fundamental layer of security in the IoT, establishing the identity and legitimacy of devices, users, processes, and other entities within the network. It involves verifying the identity claims made by these entities and allowing only authorized and trusted interactions. By implementing robust authentication protocols and mechanisms, potential risks, such as unauthorized access, data breaches, and malicious activities, can be mitigated [29].

Furthermore, authentication in the IoT goes beyond traditional methods, considering the unique challenges posed by the scale, heterogeneity, and resource constraints of IoT devices. Therefore, a lightweight and efficient authentication scheme that considers IoT devices' limited computational capabilities and storage capacities and ensures the authenticity of all entities in connected devices is always desired. A typical IoT environment with automotive services has been shown in Figure 1, in which all IoT devices lie in the device layer, connected to the IoT platform with e-commerce consumer services, and finally, the application layer, where users can access these services. The interconnected nature of devices and systems in an IoT environment with automotive services offers the delivery of improved, more secure, and more efficient use of services. However, it highlights the importance of robust safety precautions to protect against potential authentication attacks and privacy concerns linked with collecting and disseminating sensitive e-commerce electronic device data. In the bulk usage of IoT services, each entity should be authentic and legitimate before using and providing these services.

Considering the important aspect of authentication for all entities in pervasive IoT in e-commerce, in this research work, a novel scheme has been introduced, "Enabling Trust in Automotive IoT: A Lightweight Mutual Authentication Scheme for connected electronic devices in the IoT. This scheme employs a distributed architecture with a central server. Authentication is ensured and managed locally with the active participation of a central server. Within this system, gateways in the IoT network play a pivotal role as trusted entities responsible for authenticating and authorizing interactions

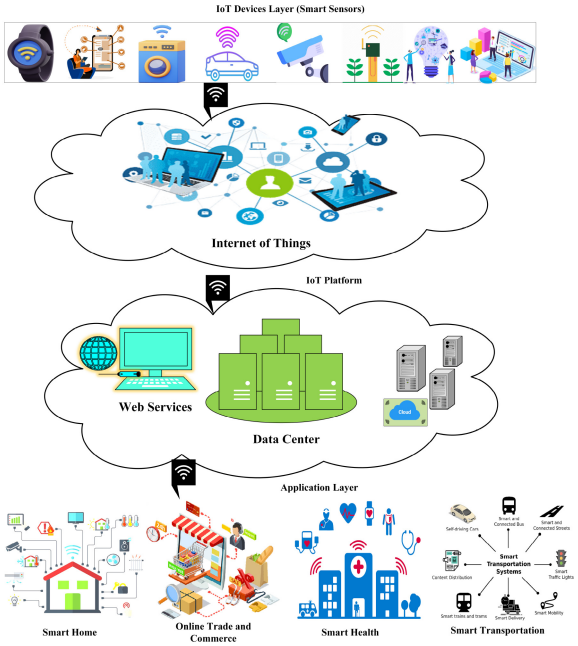


Fig. 1. A typical IoT environment with automotive access of services (devices, Gateway, Server).

within their respective vicinity. The initial phase involves devices exchanging basic messages with the gateway or central server to establish trust and verify their status. Once this initial trust is established, specific devices can communicate directly with each other without constant involvement from the central server or gateway. This direct communication is made possible because the local trust management system has authenticated and authorized these devices. Following are some of the main objectives:

- To provide a seeming-less and automatic authentication technique to authenticate all entities without the manual involvement of these entities.
- To reduce handshaking overhead and minimize congestion, significantly enhancing IoT operation.
- To optimize the authentication process and ensure effective utilization of electronic data resources without compromising security or authentication requirements.
- To facilitate authentication without the use of massive and bulky mechanisms because IoT connections need quick response and rapid query processing algorithms.

The rest of the paper is arranged as follows: Section II is Related work in Authentication schemes in IoT devices and motivation at the end of this section, Section III is the LMA scheme with all details with complete details and sequence diagram, Section IV is formal verification of LMA, Section V is based on Performance evaluation, Section VI is a comparison of LMA with other schemes, and Section VII is the conclusion of the proposed work.

II. Related Work

In the IoT context, where many devices and entities are interconnected, the authentication process can introduce additional overhead and congestion if not managed efficiently. This

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is particularly important due to the resource limitations of IoT devices, including computational capabilities and network bandwidth. Authentication of all entities in IoT infrastructures is vital for security, but it should be implemented carefully to minimize congestion and overhead. By adopting lightweight and efficient authentication mechanisms, IoT networks can maintain a secure environment while optimizing performance and resource utilization.

CAABAC [30] is one scheme that integrates contextual information with Cipher text-policy Attribute-Based Encryption (CP-ABE). The primary goal is to enhance data security and enable adaptive, contextual privacy. Regarding security, the scheme effectively addresses confidentiality, context-aware privacy, and resilience against the key escrow problem. **BCSA** [31] Introducing a novel authentication scheme based on blockchain technology. This framework combines the blockchain technique with the modular square root algorithm to establish a robust authentication process. Its validation results confirm the security and usefulness of our proposed scheme through comprehensive security analysis and detailed experimentation. **SAM** [32] Presenting an effective authentication scheme that utilizes self-adaptive and mutual key updating techniques. It assesses the performance and suitability of the solution through a comprehensive simulation that considers factors such as energy consumption, authentication failure rate, and authentication delay. It implements the proposed authentication scheme in smart home IoT systems to demonstrate feasibility and applicability. **PDA** [33] has proposed offering secure and private service advertisements and mutual authentication and has been proven to be secure according to the Canetti-Krawczyk key-exchange model. Unlike other alternatives, the proposed protocols are lightweight and require minimal modifications to existing key exchange protocols. This can be easily integrated into an open-source distributed applications framework and conducted benchmarks on various hardware platforms, including Intel Edisons, Raspberry Pis, smartphones, laptops, and desktops. **PAA** [34] explores the concept of an RFID-based IoT using widely adopted Internet protocols. Specifically, it showcases the application of the Internet Protocol Security (IPsec) protocol suite in a privacy-conscious manner. It also introduces a mutual authentication protocol that prioritizes privacy while adhering to the limitations set by the IPsec standard and conducts an analysis of its privacy and security properties. **OHE** [35] has proposed that sensitive data extracted from the IoT dataset undergo classification utilizing a Deep Learning Neural Network (DNN) structure. Data is classified, and encryption and decryption are applied only to sensitive data. The step-size Firefly method is used for key generation, and these random keys encrypt the data. Unique key generation secured the system against attacks that compromised the key generation process in IoT-based systems. **LAP** [36] is another security protocol that ensures the privacy of resource-restricted devices. Some IoT devices produce huge amounts of data where normal security mechanisms are not recommended, like hash functions and massive encryption procedures. LAP is based on lightweight encryption using only bit-wise operations. It enhances security by thwarting any malicious actions and protecting the privacy of the data. **LCAP** [37] The concept of a valid authentication period is introduced to strengthen authentication reliability between IoT devices. To decrease the time required for consecutive authentications and conserve energy during session establishment, the protocol incorporates token techniques and leverages the dynamic features of IoT devices. **LAES** [38] has proposed a scheme that reduces the number of handshakes required and effectively decreases the size of transmitted messages. For each handshake, the proposed scheme achieves a notable reduction in message size, approximately 184 bytes, compared to the 332 bytes typically observed in TLS-based schemes. The proposed scheme offers improved efficiency and optimized resource utilization, making it a promising solution for end-to-end security in communication systems. **FLAT** [39] replaces heavyweight protocols and public-key cryptographic primitives found in traditional FIdMs with lighter alternatives such as symmetric cryptographic primitives and implicit certificates. It significantly reduces data exchange overhead by approximately 31% compared to a baseline solution. Moreover, the FLAT client, representing IoT devices in the protocol, outperforms the baseline client regarding data exchange, storage, memory, and computation time. **ELA** [40] involves a registration center that generates public and secret information for each node during network entry. Once registered, the protocol operates independently, enabling nodes to perform mutual authentication, secure key exchange, and communications without relying on the registration center. It reduces computational complexity and overhead, addressing potential attacks while ensuring secure operations in the IoT environment. **LEA** [41] has proposed a lightweight and escrow-free scheme focusing on security and performance regarding energy consumption and processing time on the TelosB platform. It highlights the flaws in certain highly efficient schemes and demonstrates that combining SMQV with implicit certificates results in secure and lightweight authentication. **M2M-CRC** [42] utilizes lightweight operations such as xor, addition, subtraction, and a hash function. Notably, it achieves authentication with only four exchanged messages between the parties involved, making it efficient in execution time and communication cost. The security of the protocol was rigorously evaluated using formal methods like *Burrows-Abadi-Needham* (BAN) *logic and Automated Validation of Internet Security Protocols and Applications* (AVISPA) tool, supplemented by an informal analysis of its resilience against known attacks. **LCDMA** [43] replaces the high-complexity bilinear pairing typically found in traditional schemes with a symmetric polynomial. Theoretical analysis conducted under the random oracle model demonstrates that LCDMA not only withstands common attacks but also ensures secure traceability while maintaining anonymity. Furthermore, performance evaluation reveals that the proposed scheme outperforms existing representative schemes regarding computation and communication overhead. **LAKS-NVT** [44] eliminates the need for a server verification table, making it more efficient and lightweight. It thoroughly evaluates the security of LAKS-NVT against various potential attacks. It demonstrates its effectiveness in achieving secure mutual authentication between sensor nodes

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TABLE I
LITERATURE SUMMARY

Scheme Name	Parameters	Main Idea	Disadvantages
CAABAC [31]	Contextual, CP-ABE	Adaptive Contextual Privacy, Key Escrow	Congestion, Delay
BCSA [32]	Blockchain, Modular Root	Robust authentication, comprehensive security analysis	latency, overflow, cost
SAM [33]	Self-adaptive key updating	Energy efficiency, low authentication failure rate	Delay, less secure
PDA [34]	Lightweight, Easy integration	Privacy-conscious, minimal modifications in protocols	resource needed, cost
PAA [35]	RFID-based IoT, IPsec, MA	Privacy-conscious, adherence to IPsec limitations	time consuming, resources needed, IP sec based
OHE [36]	Deep Learning, SFF optimization	Enhanced privacy for IoT data, robust encryption	Delay in auth, conventional mechanisms
LAP [37]	XOR, anti-counterfeiting	Improved security, lightweight encryption	processing cost, delay
LCAP [38]	Valid authentication, tokens	Enhanced robustness, efficient, energy-saving	low security, token compromise
LAES [39]	Reduced handshakes, messages	Improved efficiency, optimized resource utilization	time consuming, low security
FLAT [40]	Lightweight, symmetric, Certificates	Reduced exchange overhead, improved performance	based on conventional security protocols
ELA [41]	Registration center, reduced complexity	Reduced complexity, secure operations	congested, latency
LEA [42]	escrow-free, SMQV implicit certificates	Secure and lightweight, performance improvement	need more calculations, complex
M2M-CRC [43]	XOR operations, hash function	Efficient authentication, low execution time, cost	memory and communication costs
LCDMA [44]	Symmetric polynomial, anonymity	Withstands attacks, improved computation overhead	mathematical and complex
LAKS-NVT [45]	Eliminates server verification table, mutual authentication, session key security, efficiency, lightweight, server-based, congested, sessions keys can be compromised		

and servers using Burrows-Abadi-Needham logic. The well-known Real-Or-Random model further supports the security analysis, proving that LAKS-NVT provides session key security. All these schemes are summarized in Table I, including their key principles, parameters employed in securing the authentication process, and their basic limitations.

RAFI [45] is proposed for handling IoT-centric RFID infrastructure security. It uses a random oracle technique for secure authentication of devices in IoT. It works better regarding results, but it creates more traffic and congestion due to the massive use of messages. MHE-IS-CPMT [45] used the concept of ECC with secure transmission of keys and authenticated the data exchange process. It uses five steps, manages all the processes with session keys, and updates them. The home network head registers users and devices, and the ECC authenticates each entity. It proved to be a better security measure against different types of attacks. **SLAS** [46] has proposed a modified honey encryption for 6G/IoT devices. This scheme uses the main ideas of Kaul and Awasthi and modifies them. The results proved better secure authentication with a minimum cost of resources. It is more useful in 6G scenarios, but if truly implanted, many technical issues are there. **LEA-PSC** [38] is based on end-to-end authentication for IoT and is recommended for minimizing message overhead in communication using a symmetric-key encryption technique. TLS-based authentication is preserved, which makes authentication easy with minimum bandwidth utilization. It also reduces the control of the message of handshaking and broadcasting. **SES** [47] used SET and 3D Secure for secure authentication in IoT based. These techniques are fundamental for e-commerce services with less computation and communication costs. It uses analytical procedures and extensive simulations to prove the authenticity of services provided by e-commerce. Table I summarizes all these schemes, including their key principles, parameters for securing the authentication process, and their basic limitations.

A. Motivation

All of the aforementioned schemes have ensured the authentication of entities, including services, users, and processes. However, these schemes have used massive and heavy authentication techniques like RFIDs, biometric authentication, Multi-factor authentication, User IDs, and hybrid authentication. These schemes create additional overhead and congestion, which is time-consuming and requires some extra procedures for authentication. In the case of IoT-connected devices, this extra burden on the system can hinder the overall performance and efficiency in delivering these pervasive services. Consequently, there is a need for a novel authentication scheme that can automatically authenticate all entities with minimal congestion and also

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ensure efficient resource management. In the context of connected electronic devices within a pervasive environment, the objectives are linked to refining the authentication process while ensuring optimal use of network resources. The main aim is to provide a lightweight and secure authentication mechanism without sacrificing the efficiency required for authentication.

III. Lightweight Mutual Authentication Scheme (LMA)

The newly proposed scheme provides an improved and automatic authentication process for all entities involved in the IoT environment. Its primary goal is to ensure the authentication of all entities involved in communication within its surroundings. This scheme makes the system more robust against unauthorized actions against IoT infrastructure, especially in e-commerce. The authentication process is automotive because it takes place in the background without manual authentication, and all the entities automatically authenticate each other. For e-commerce services using IoT, users and devices do not need to verify at each step; otherwise, too much handshaking and repetitive authentication procedures degrade system performance. It satisfies end users without knowing the background process, and any service can be used if it is legitimate and authentic. LMA uses a mechanism that allows user credentials to be exposed under any conditions to ensure anonymity. The system verifies each user or device, even the server and gateway, to ensure that any entity involved in the transaction is verified and authenticated before being involved in electronic transactions. Any electronic service that can not ensure user data privacy will make users reluctant to use it. LMA is lightweight and has a more secure position compared to other services. Using unlinkability, the system never reveals the transaction behavior, and tracking online purchase orders is also an essential aspect of electronic IoT. LMA ensures these aspects by applying different mechanisms in the background, mitigating possible threats like identity theft and fraudulent authentications. No attacker can intercept the user's sensitive data, and it thwarts any attempt by a malicious user.

A. IoT Connected Network Model of LMA

Consider an IoT network, denoted as *NT*, where each device is referred to as nodes (multiple nodes) N_n (where $n = 1, 2, 3, \ldots, N$) are connected to gateways GW_{sn}. These gateways are then connected to a central server *SVR* for further processing and execution of various tasks. The intelligent devices in this network cooperate and provide services based on their connections. The distance between devices/nodes, represented as DT_{ij}, indicates the distance between any two devices along the x and y axes. These devices are closely connected in a hybrid and heterogeneous network, allowing them to sense each other directly. The connectivity between devices is determined by the condition SR_{ij} ! DT_{ij}, establishing a connected IoT environment. In this hybrid IoT network, each device is directly connected to a gateway, while all gateways are connected to central servers, and service users interact with the central server. This connectivity is represented as PH_{ij}, where $0 \leq PH \leq N(N-1)/2$ and $PH = N(N-1)^2$. Table II presents the symbols used in this context and their corresponding meanings.

TABLE II
Symbol's Table

Symbols	Meaning
SVR_{key}	Super key generated by Server
S_n	Node/device
ID_{sn}	Unique ID of each node/device
USR_n	Service user
ID_{USR}	Unique ID of service user
GW_{sn}	Gateway
ID_{GW}	Identity of Gateway
K_n	Session key for each Session
DT_{ij}	Distance between Devices(sensors)
PH_{ij}	Connectivity of all devices
Tn	Time Stamps and nounces
Cer_{sn}	Certificate of Gateway
PR_{Svr}	Private key of Server
PU_{sn}	Public Key of Any Device
PU_{USR}	Public Key of service user
PR_{svr}	Private Key of Server
En	Encryption Process
DN	Decryption Process
T_{exp}	Time of Expiry

B. System Model and Working Procedure of LMA

Step-1: The server creates a super secret key SVR_{key} and each S_n has creates a unique key ID_{sn} while every service user USR_N has also creates its unique key ID_{USR}. There are many Gateways GW_{sn} in the network *NT*, and each GW_{sn} has a unique ID_{GW}. Now, any S_n can send its request to server *Server* through GW_{sn} and in the same way, each USR_n can send its request to server *Server* through GW_{sn} for connection. The S_n first calculates the Hash of the time-stamped and its identity and then is forwarded to GW_{sn}. We expressed this process as $S_n \rightarrow GW_{sn}$ while $H[ID_{sn} \bigoplus K_n]$. The GW_{sn} receives all these parameters,

$$S_n \rightarrow GW_{sn}\left(H\left[ID_{sn} \bigoplus K_n\right] + T1\right) \tag{1}$$

While each user USR_n also calculates the Hash of the time-stamped and its identity and forwards to GW_{sn}. This can be expressed as $USR_n \rightarrow GW_{sn}$ while $H[ID_{USR} \bigoplus K_n]$. The GW_{sn} receives all these parameters for any user as,

$$USR_n \rightarrow GW_{sn}\left(H\left[ID_{USR} \bigoplus K_n\right] + T1_n\right) \tag{2}$$

Step-2: The Gateway verifies the contents of the received messages from the hash of the message and forwards the same messages by concatenating their identities.

$$GW_{sn} \rightarrow S_n\left(H\left[ID_{sn} \bigoplus K_n\right] + T1 || \left[ID_{GW} + T1\,'\right]\right) \tag{3}$$

$$GW_{USR} \rightarrow USR_n\left(H\left[ID_{USR} \bigoplus K_n\right] + T1 || \left[ID_{GW} + T1\,'\right]\right) \tag{4}$$

Upon receiving a message from the gateway, the server undertakes a series of actions to ensure the message's integrity and authenticity. Firstly, it calculates the time needed to get the messages. The delay factor is calculated from the current time

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and time stamped inside these messages. The calculated delay is measured and compared with a threshold; all those packets are dropped, which feels more delayed than the threshold. Message correction is checked by calculating the hash of the data and verifying if any impersonation is dropped. The server further verified the message's authenticity by verifying with the Gateway. When the message is verified not to be tempered during transmission, the concerned device confirms it. All this guarantees better security and enhanced privacy for sensitive data.

Step-3: The server creates a response message after confirming the integrity of the message with Gateway and the device. This response message has information like public key certificates and timestamps. The certificate is the digital identity of the server, and the timestamp is the accurate time when the message was created. Using these two things (public key certificate and timestamp), the server ensures the security and integrity of the response message. In this case, the timestamp is used to keep audit records in proper order, while the public certificate authenticates that the server is legitimate.

$$S_n \rightarrow GW_{sn}([Cer_{sn}]||T2) \tag{5}$$

For the user, the message will be,

$$S_n \rightarrow GW_{sn}([Cer_{sn}]||T2_n) \tag{6}$$

The certificate is a digital document for server verification and can be used by any node or Client user.

$$Cer_{sn} = En\big(PR_{Svr}, \big[T_n||ID_{sn}||PU_{sn}||T_{exp}\big]\big) \tag{7}$$

In the same way, for each user,

$$Cer_{sn} = En\big(PR_{Svr}, \big[T_n||ID_{USR}||PU_{USR}||T_{exp}\big]\big) \tag{8}$$

Step-4: Any node/user can verify the identity of the Server by decrypting the same message with the public key of the Server.

$$\begin{aligned} &Dn(PU_{Svr}, Cer_{sn}) \\ &= Dn\big(PU_{svr}, En\big(PR_{svr}, \big[T2||ID_{sn}||PU_{sn}||T_{exp}\big]\big)\big) \\ &= [T2||ID_{sn}||PU_{sn}||T_{exp}] \end{aligned} \tag{9}$$

While any user can verify the server by decrypting the same message with the public key of the Server,

$$\begin{aligned} &Dn(PU_{Svr}, Cer_{USR}) \\ &= Dn\big(PU_{svr}, En\big(PR_{svr}, \big[T2_n||ID_{USR}||PU_{USR}||T_{exp}\big]\big)\big) \\ &= \big[T2||ID_{USR}||PU_{USR}||T_{exp}\big] \end{aligned} \tag{10}$$

In the same way, any other device can communicate with the Server and verify the Server's authenticity. For any other device, communicating with the server is like,

$$Cer_{sn} = En\big(PR_{Svr}, \big[T_n||ID_1||PU_1||T_{exp}\big]\big) \tag{11}$$

It can be decrypted, like,

$$\begin{aligned} &Dn(PU_{Svr}, Cer_1) \\ &= Dn\big(PU_{svr}, En\big(PR_{svr}, \big[T2||ID_1||PU_1||T_{exp}\big]\big)\big) \\ &= \big[T2||ID_1||PU_1||T_{exp}\big] \end{aligned} \tag{12}$$

In the same way, other devices, like

$$\begin{aligned} &Dn(PU_{Svr}, Cer_2) \\ &= Dn\big(PU_{svr}, En\big(PR_{svr}, \big[T2||ID_2||PU_2||T_{exp}\big]\big)\big) \\ &= \big[T2||ID_2||PU_2||T_{exp}\big] \end{aligned} \tag{13}$$

In general form for all "N" number of devices like,

$$\begin{aligned} &Dn(PU_{Svr}, Cer_{n+1}) \\ &= Dn\big(PU_{svr}, En\big(PR_{svr}, \big[T2||ID_{n+1}||PU_{n+1}||T_{exp}\big]\big)\big) \\ &= \big[T2||ID_{n+1}||PU_{n+1}||T_{exp}\big] \end{aligned} \tag{14}$$

Similarly, any Client user can communicate with the server and validate its authenticity. For any other user, communicating with the server is like,

$$Cer_{USR} = En\big(PR_{Svr}, \big[T_{n+1}||ID_{1+n}||PU_{1+n}||T_{exp}\big]\big) \tag{15}$$

The decryption will be,

$$\begin{aligned} &Dn(PU_{Svr}, Cer_{1+n}) \\ &= Dn\big(PU_{svr}, En\big(PR_{svr}, \big[T2+n||ID_{1+n}||PU_{1+n}||T_{exp}\big]\big)\big) \\ &= \big[T2+n||ID_{1+n}||PU_{1+n}||T_{exp}\big] \end{aligned} \tag{16}$$

For all other users,

$$\begin{aligned} &Dn(PU_{Svr}, Cer_{2+n}) \\ &= Dn\big(PU_{svr}, En\big(PR_{svr}, \big[T2+n||ID_{2+n}||PU_{2+n}||T_{exp}\big]\big)\big) \\ &= \big[T2||ID_{2+n}||PU_{2+n}||T_{exp}\big] \end{aligned} \tag{17}$$

While for generic form for all "n" number of users,

$$\begin{aligned} &Dn\big(PU_{Svr}, Cer_{n'+1}\big) \\ &= Dn\big(PU_{svr}, En\big(PR_{svr}, \big[T2||ID_{n'+1}||PU_{n'+1}||T_{exp}\big]\big)\big) \\ &= \big[T2||ID_{n'+1}||PU_{n'+1}||T_{exp}\big] \end{aligned} \tag{18}$$

The server in this IoT architecture possesses comprehensive knowledge about all the devices within the network, including their functionalities and services. Acting as a centralized authority, the server is important in managing the traffic flow and directing communication within the IoT network. It serves as a central hub through which all communication passes. Initially, the server establishes a foundation of trust by exchanging important initial messages with the devices, users, and the gateway. These messages often involve authentication and verification processes to ensure the devices, users, and gateway are legitimate entities within the IoT network.

Once the initial trust is established, the Server relinquishes direct control over the communication between the devices, users, and the Gateway. It grants autonomy to the devices and allows them to communicate directly with each other or with the users or gateway without constant intervention from the Server. This decentralized approach enables efficient and direct communication between the entities in the network. However, while the Server in the background actively manages the communication, it plays a vital role in adjusting traffic and monitoring the network. It manages the overall network operation, ensuring communication flows smoothly and efficiently. The Server may also act as a mediator or facilitator if conflicts or issues arise during communication between devices, users, or the Gateway.

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Step-5: After the third message from Device-1 to the Server, a specific action is triggered. The Server responds to Device-1 by encrypting the same request, including the public key of Device-2 and, in the same way, to any user, like user-1, in the response. This response serves a purpose: it enables direct communication between Device-1 and Device-2/user-1 without requiring constant involvement from the Server. By including the encrypted request and the public key of Device-2/user-1 in the response, the Server facilitates a direct and secure channel between the two devices or users. This allows Device-1/user-1 to directly communicate with Device-2/user-2, reducing the reliance on the Server as an intermediary. Consequently, communication between the devices becomes more efficient as it bypasses the additional steps and potential delays involved in routing messages through the Server.

$$Svr \rightarrow S_n(En(PR_{svr}[PU_{svr}||T2])) \tag{19}$$

For any client users

$$Svr \rightarrow USR_n(En(PR_{svr}[PU_{svr}||T2])) \tag{20}$$

Upon receiving the message, Device-1 gains important information about Device-2 or user-1/user-2 about device-1/device-2, including its identity (represented as PU_2, the type of services it offers, and additional details. This information is encrypted using the private key of the Server (PR_{svr}) to ensure confidentiality and security. To access and understand the encrypted information, Device-1 employs the public key of the Server (PU_{svr}) to decrypt the message. Device-1/user-1 can securely decipher the encrypted data and obtain the relevant information by utilizing the appropriate key. Once decrypted, Device-1/user-1 knows about other network entities or devices that provide the desired service. This information allows Device-1/user-1 to identify and locate other resources that offer similar services. Moreover, Device-1/user-1 gains clarity regarding the access and authorization requirements for these resources, as determined by the Server. This process empowers Device-1 to make informed decisions regarding resource utilization and access. By decrypting the message using the public key of the Server, Device-1 gains valuable insights into the network's ecosystem and the availability of services from various entities. This knowledge facilitates efficient resource allocation and access management, enabling Device-1/user-1 to make appropriate requests and interact with the network securely and authorized.

Step-6: Having analyzed and decrypted the aforementioned messages, Device-1/user-1 gains the necessary information to establish direct communication with Device-2/user-2. With this knowledge, Device-1/user-1 can initiate a message directly to Device-2/user-2 without the Server's involvement. By bypassing the Server and communicating directly, Device-1/user-1 can expedite the transmission process and reduce potential delays. The direct communication pathway allows for a more efficient and streamlined exchange of information between Device-1 and Device-2 (user-1, Device-1, Device-2). This direct message transmission enhances the responsiveness and agility of the communication process. Device-1/user-1 can transmit its message directly to Device-2, eliminating the additional steps and potential latency associated with routing the message through the server. The direct communication between Device-1/user-1 and Device-2 fosters a more efficient and prompt data exchange, facilitating quicker responses and reducing overall communication overhead. It enables Device-1/user-1 to directly interact with Device-2 based on the information obtained from the decrypted message without requiring continuous involvement or oversight from the server.

$$S_1 \rightarrow S_2(En(PU_2[ID_1||N1]) + Cer_1) \tag{21}$$

$$USR_1 \rightarrow S_2(En(PU_2[ID_1||N1]) + Cer_1) \tag{22}$$

$$S_1 \rightarrow USR_2(En(PU_2[ID_1||N1]) + Cer_1) \tag{23}$$

Upon receiving the message, Device-2/User-2 is presented with several components. These include Cer_1 (Certificate of Device-1), ID_1 (identity of Device-1/user-1), and a unique nonce N_1. Using the public key certificate of Device-2 or User-2 PU_2, the ID_1 and the N_1 are encrypted. Nonces in each transaction are embedded to confirm their uniqueness. Each part of the message is important, and each one plays an important role. First, it authenticates the Device-1/user-1 from Cer_1 and ID_1. The server also verifies this authenticity, and it is proven that the requesting device or user is a legitimate one. The message's nonce N_1 prevents many attacks, like man-in-the-middle and replay. These nonce's uniquely identify each message with a unique transaction identity, and they also detect if any attacker used previous or replayed current messages.

The public key PU_2 ensures the confidentiality of the message by encrypting ID_1 and N_1, and it demands decryption by the indented recipient, which may be Device-2 or User-2. When Device-2 receives this message from Device-1, it asks the server to verify its authenticity. The server verifies Device-1 by checking Cer_1 and ID_1 with Device-2. Currently, public keys PU_n of Device-1 or user-1 and Device-2 or user-2 are delivered secretly, and a secure mechanism on both sides is developed. Furthermore, the identities of each device or user are securely exchanged on both sides, which makes it possible to start secure communication. However, additional measures are needed to secure the communication further, ensure a robust trust level, and thwart any interception or man-in-the-middle attacks.

Step-7: Device-1 or user-1 sends a request, and Device-2 or user-2 responds with a message. This message now has two nonces, $N1$, and a fresh nonce, $N2$, and is encrypted by the same device public key (PU1). Embedding nonce $N1$, clearly verifies the message is coming from Device-2 or user-2, which is necessary for authenticating the sending side. It also acknowledges the trust-building between sender and receiver. Now Device-2 uses the public key of Device-1 or user-1 ($PU1$) to encrypt the message with nonces $N1$ and $N2$. The message is ready to be sent to Device-1 or User-1. Embedding these nonces $N1$ and $N2$ is useful in building trust between two devices or users and is necessary for further verifying the communication partners.

$$S_2 \rightarrow S_1(En(PU_1[N1||N2])) \tag{24}$$

$$USR_2 \rightarrow S_1(En(PU_1[N1||N2])) \tag{25}$$

$$USR_2 \rightarrow USR_1(En(PU_1[N1||N2])) \tag{26}$$

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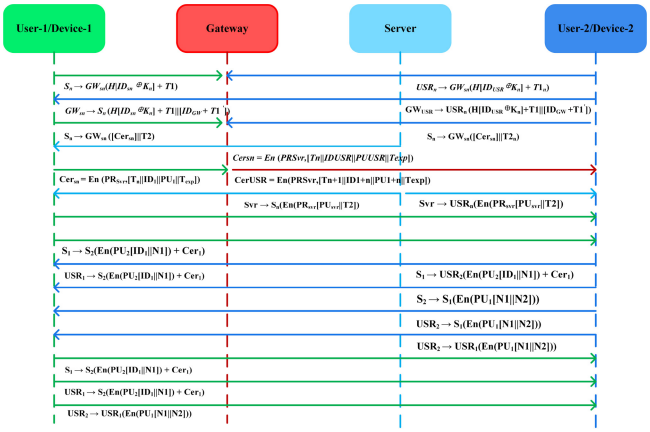


Fig. 2. LMA Sequential Flow of Messages between Users, devices, gateway and server.

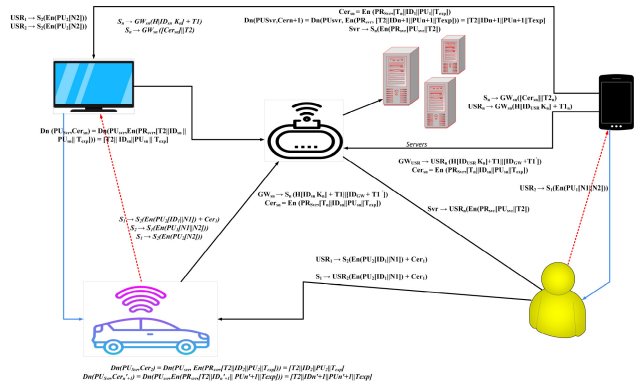


Fig. 3. LMA process between two Devices, Gateway and Server.

Step-8: Device-1 or User-1 uses its private key, (PR_1"), to decrypt this encrypted message after receiving the response from Device-2 or User-2. The nonces inside the encrypted message ensure thwart replay attacks. At this stage, before using Device-2's services, it sends back a message. This message contains an encrypted nonce ($N2$) with a public key of Device-2 ($PU2$). This further verifies the message's authenticity, indicating that Device-1 or User-1 has sent it. This step further authenticates the trust between two sides, either between devices or users.

$$S_1 \rightarrow S_2(En(PU_2[N2])) \tag{27}$$

$$USR_1 \rightarrow S_2(En(PU_2[N2])) \tag{28}$$

$$USR_2 \rightarrow S_2(En(PU_2[N2])) \tag{29}$$

Therefore, a total of seventeen messages are necessary to achieve a comprehensive understanding and confirmation of messages exchanged between the devices and users. However, the initial eight messages are infrequently used since, upon receipt, the nodes store the public keys for future reference. This technique, commonly called "caching," allows the devices to retain and utilize the public keys without repeated exchanges. All these steps from Devices-01/user-1 to Device-02/user-2, Gateway, and Server have been shown in Figure 2. The flow diagram of all these actions between Sever, Gateway, Devices, and Users is shown in Figure 3.

Algorithm 1: LMA

Input: Server creates a super secret key *SVRkey* and each *Sn* has a unique *IDsn*. There are many Gateways *GWsn* in the network *NT* and each *GWsn* has a unique *IDGW*.

// Step 1
foreach $Sn \rightarrow GW_{sn}(H[ID_{sn} \oplus K_n] + T1)$ **do**
end
foreach $USR_n \rightarrow GW_{sn}(H[ID_{sn} \oplus K_n] + T1)$ **do**
end

// Step 2
foreach $GW_{sn} \rightarrow Sn(H[ID_{sn} \oplus K_n] + T1||[ID_{GW} + T1'])$ **do**
end
foreach $GW_{sn} \rightarrow Sn(H[ID_{usr} \oplus K_n] + T1||[ID_{GW} + T1'])$ **do**
end

// Step 3
foreach $Sn \rightarrow GW_{sn}([Cer_{sn}]||T2)$ **do**
  $Cer_{sn} = En(PR_{svr}, [Tn||ID_{sn}||PU_{sn}||T_{exp}])$
end
foreach $USR \rightarrow GW_{USR}([Cer_{USR}]||T2)$ **do**
  $Cer_{USR} = En(PR_{svr}, [Tn||ID_{USR}||PU_{USR}||T_{exp}])$
end

// Step 4
foreach *Node/Device* **do**
  $Dn(PU_{Svr}, Cer_{sn}) =$
   $Dn(PU_{Svr}, En(PR_{svr}, [T2||ID_{sn}||PU_{sn}||T_{exp}])) =$
   $[T2||ID_{sn}||PU_{sn}||T_{exp}]$
end
foreach *Node/Device* **do**
  $Dn(PU_{Svr}, Cer_{USR}) =$
   $Dn(PU_{Svr}, En(PR_{svr}, [T2||ID_{USR}||PU_{USR}||T_{exp}])) =$
   $[T2||ID_{USR}||PU_{USR}||T_{exp}]$
end

// Step 5
foreach *Sn* **do**
  $Svr \rightarrow Sn(En(PR_{svr}, [PU_{svr}||T2]))$
end
foreach *USR* **do**
  $Svr \rightarrow USR(En(PR_{svr}, [PU_{svr}||T2]))$
end

// Step 6
$S1 \rightarrow S2(En(PU2[ID1||N1]) + Cer1)$
$USR1 \rightarrow S2(En(PU2[ID1||N1]) + Cer1)$

// Step 7
$S2 \rightarrow S2(En(PU1[N1||N2]))$ $USR1 \rightarrow S2(En(PU1[N1||N2]))$

// Step 8
$S1 \rightarrow S2(En(PU2[N2]))$ $USR1 \rightarrow S2(En(PU2[N2]))$

In the next section, the LMA Algorithm 1 has been defined. The LMA algorithm demonstrates a step-by-step process of meticulously defining each step with appropriate notations. It begins with initialization, where essential variables and data structures are set up, and communication channels are established.

IV. Formal Verification of LMA

This section checks the proposed Lightweight Mutual Authentication scheme using formal methods. We will define the properties in the CTL model checker and verify their validity.

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TABLE III
SIMULATION PARAMETERS

Parameter	Values
Simulation Time	1000 Seconds
Maximum Sensors	100
Sensors (devices) in each row	20
Payload size	100 Bytes
Distance B/W pair od sensors	10-15 meters
Distance of GW	50-120 meters
Uplink resource for NB-IoT	Single-tone with 15 kHz, 5 RUs
Downlink resource for NB-IoT	1 PRB, 5 SFs
Power saving technique	PSM
T3324 timer	30 seconds
T3412 timer	1 hours

A. CTL Model Checking

To perform CTL model [48] checking, we define the CTL formulas that express the properties we want to verify for the LMA scheme.

Let's consider the following CTL formulas: φ_1 : AG($S1 \rightarrow$ AF($S2 \rightarrow$ EG($Svr \rightarrow$ AF($S2 \rightarrow$ A[¬Cer1]U Cer1))))
φ_2 : AG($S2 \rightarrow$ A[¬Cer1]U Cer1)

Now, let's check the validity of these properties for the LMA scheme.

1) Property 1 (φ_1): Property 1 states that it is always globally true that if $S1$ holds, then it is always in the future true that if $S2$ holds, then eventually, in all future states where Svr holds, it is always in the future true that if $S2$ holds, then there exists a path where Cer1 is false until Cer1 becomes true.

Model satisfies Property 1.

2) Property 2 (φ_2): Property 2 states that it is always globally true that if $S2$ holds, a path exists where Cer1 is false until Cer1 becomes true.

Model satisfies Property 2.

Therefore, based on the CTL model checking, we can conclude that the LMA scheme satisfies both properties.

V. PERFORMANCE EVALUATION OF LMA

The performance evaluation of Lightweight Mutual Authentication involves assessing various aspects related to the efficiency and effectiveness of the authentication scheme. Table III outlines the specified parameters and their respective values utilized in simulations.

A. Packet Delivery Ratio in LMA

The Packet Delivery Ratio (PDR) is a metric calculated by dividing the number of successfully received packets by the total number of packets sent by the sending device. It serves as a crucial indicator for assessing IoT network performance. Here, we have mapped the system behavior in Figure 4, and it is clear that as the number of devices increases, the PDR rate tends to decline due to the congestion within the IoT network. This decrease in PDR can be attributed to the increased traffic and communication load caused by more devices. It has been observed that LMA has a minor impact on system performance in terms of PDR. Without LMA, the system exhibits slightly better performance with an increased PDR. However, after implementing LMA, there is a slight decrease in PDR. For instance, at the beginning of the simulation, without LMA, the PDR is measured at 80%, whereas with LMA, it is slightly lower at 79.87%. As the number of devices increases up to 100, the PDR without LMA declines to 70%, and with LMA, it decreases to 69%. Despite these minor decreases, LMA still demonstrates its effectiveness in maintaining system performance while providing the benefits of mutual authentication. The decrease in PDR associated with LMA implementation is minimal, indicating that LMA effectively maintains the system's overall performance.

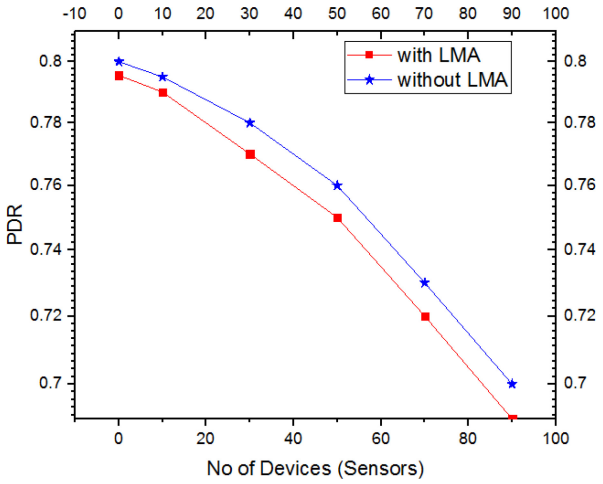


Fig. 4. Packet Delivery Ratio (PDR) % versus No. of Devices with LMA & without LMA.

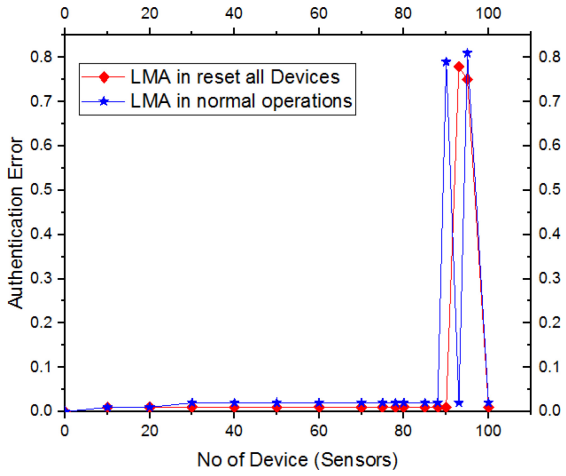


Fig. 5. Authentication Error in 100 Device in IoT network.

B. Authentication Error Rate

It has been observed that there is an increase in authentication errors and delays when many devices attempt to authenticate with a server or gateway simultaneously. In Figure 5, a spike in the graph indicates a significant increase in authentication errors as the number of devices increases to 100. This spike suggests that the authentication system may

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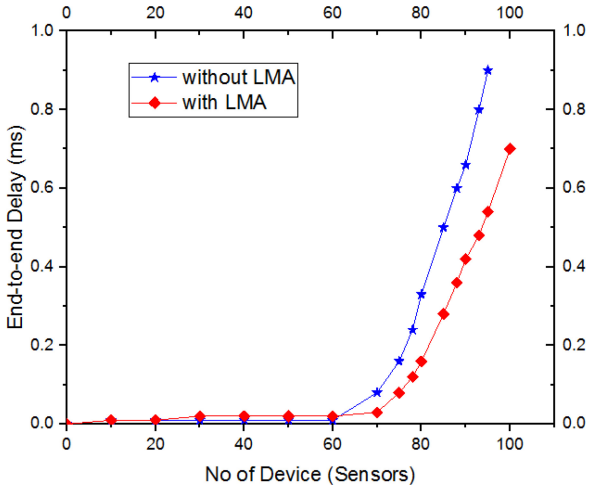


Fig. 6. End to End Delay in ms in LMA compared without LMA.

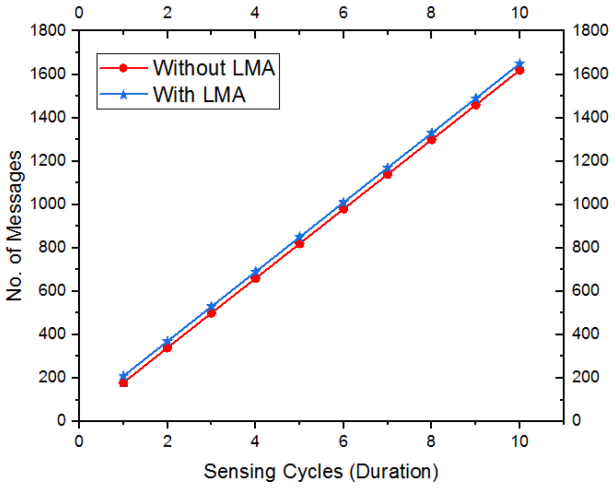


Fig. 7. No. of Messages in each Sensing Cycle.

struggle to handle the increased load and process authentication requests efficiently. As more and more devices try to authenticate simultaneously, the system may struggle to handle all the requests. This can be due to factors such as limited processing power, network congestion, and the availability of authentication certificates. These issues can lead to delays in processing authentication requests and may require the devices to resend requests or request new certificates after a specific threshold.

C. End to End Delay in LMA

There is a noticeable trend toward an increase in end-to-end delay (E2ED) with the increasing number of devices. This relationship can be attributed to two main factors: network congestion and the overall distance covered by the IoT network. Based on the experiment and Figure 6, it has been observed that the system experiences less delay when the LMA is not present. However, when the LMA is introduced, there is a slight increase in delay as the number of devices increases.

When fewer nodes are at the beginning of the simulations, the system operates smoothly without any noticeable delay, and data is transmitted promptly. However, as the number of devices increases to around 60, delays emerge in both scenarios, with and without LMA. The delay increases exponentially once the number of devices reaches approximately 70 nodes. This suggests that the system becomes more congested or faces limitations in processing capacity as the number of requests for authentication increases. Consequently, the delay reaches its maximum point with 100 devices in the IoT network. Overall, these findings indicate that the presence of an LMA introduces a slight increase in delay compared to the scenario without an LMA, especially as the number of devices exceeds a certain threshold. This could be attributed to the additional overhead and processing requirements associated with the LMA's role in managing and coordinating the mobility of the devices, as shown in Figure 6.

D. Number of Messages

The system has also tested for several messages with increasing simulation rounds. When the LMA is in place, it processes the information and sends it to other devices and the central center. With the increasing number of cycles (time and process), there is a gradual increase in the number of messages, while without LMA, the system performs the same with little decrease in sending packets. In these processes, messages are transmitted more frequently. If there is no filtering and switching, then the LMA provides even minor changes in data that might trigger messages. This may conclude that the presence of an LMA can help optimize the messaging process during sensing cycles by applying a security policy. At the same time, without an LMA, the number of messages might more closely follow the patterns, leading to a higher number of messages, as shown in Figure 7. For these experiments, it is important that the scenario is specifically designed for an increasing number of messages, and the main task was to manage and optimize data processing and messaging. The actual behavior could be changed and vary depending on the specific context in IoT environments. There is a total of 8.5% change in these values if LMA exists.

E. Reconfiguration Time in LMA

The IoT environment is tested and checked for reconfiguration and authentication at various times with sensing cycles. All these devices and objects are used for collecting and exchanging information over the Internet. Reconfiguration refers to making changes to the arrangement and setup of the objects within an IoT environment. Here, authentication of these objects after reconfiguration is used to optimize the security process and its effect on overall system performance. An increasing number of objects may lead to more data being generated, transmitted, and processed throughout the IoT network. The relationship between the increasing number of devices and the time it takes for reconfiguration is mapped in Figure 8. Here, the experiments range from 1 second to 15 seconds, with an increasing number of devices. With the increasing complexity of managing more objects, the time

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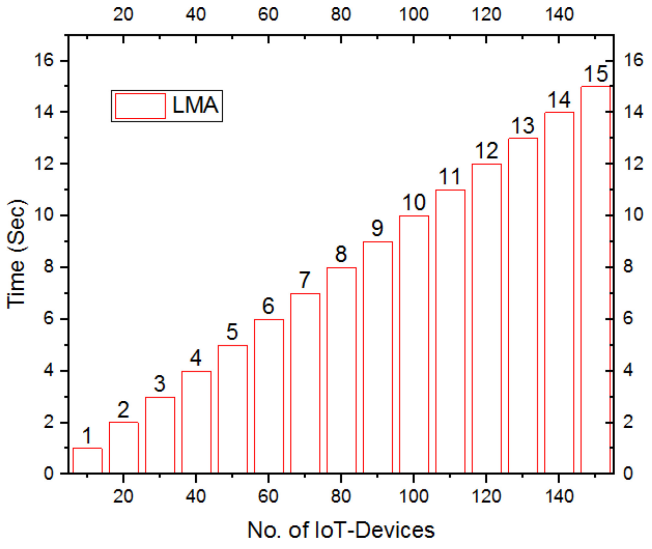


Fig. 8. Reconfiguration Time in Seconds.

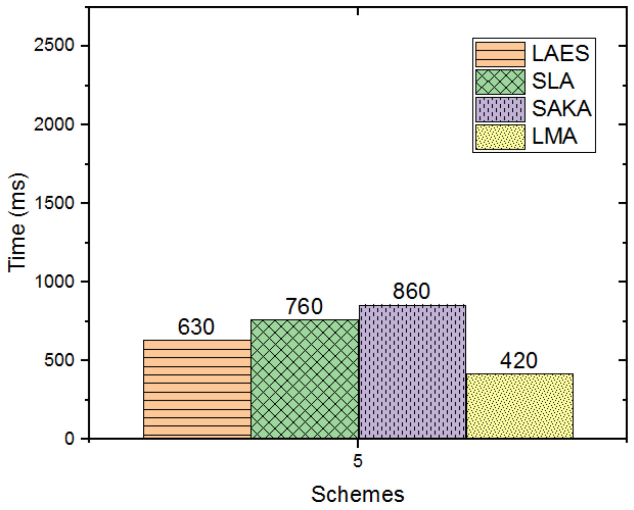


Fig. 9. LMA Computation Cost comparison with other Scheme.

required for the mentioned process increases to 15 seconds because of an increasing number of authentication processes. It is also clear from these experiments that as the IoT network expands and the number of devices grows, there are more authentication attempts. There is some overhead in authentication if there is frequent reconfiguration.

VI. Comparison With Other Schemes

Although the LMA is checked and evaluated using parameters using some dominant metric values, this section compares it with other state-of-the-art schemes to check its applicability in a real-time scenario.

A. Computation Cost

The computation cost of LMA is compared with the other three schemes [45], [46], and [38] shown in Figure 9. Computational time is the time it takes to handle a request and complete the data retrieval process. LMA cost is minimal compared to these other schemes due to the authentication mechanism and its lightweightness. We have equally mapped LMA with other schemes in the same scenario and mapped the results into a graph in which the schemes are numbered on the x-axis and the y-axis, and there is time (ms), as shown in Figure 9. The first scheme [45] has 630 ms, scheme [46] 760 ms, and scheme [38] has 860 ms, respectively, while LMA has a maximum of only 420 (ms) of time-consuming time in the authentication process. Lower computation cost means that it utilized less time for authentication and proved that this scheme is lightweight and automatic compared to others. These other schemes involve some extensive security techniques like RFID, hash functions, and bit-wised operations, while we have used secure key authentication using the concept of distributed vital operations. Fewer messages are communicated between the server, gateway, users, and devices for authentication checks and data validity between these entities. These other schemes provide more layers of security for IoT-based electronic devices, which need more time and extensive data processing. Calculating the average of all these three other schemes is 750 ms, and LMA stands at 420 ms, so there is a 56% decrease in the computation cost.

B. Communication Cost

Communication is calculated from user/device requests, gateway messages, and server responses. These entities transmit different bit levels. Using these bits, we have calculated the communication cost. The LMA is compared with other schemes, including [45], [46], and [49], shown in Figure 10, to find out the communication cost. All the other three schemes have also been experimented with in the same scenario as set for LMA, and simulations for each one of them have run. These values are mapped with the x-axis representing the schemes, and on the y-axis, the number of bits for the set completing these experiments is shown in Figure 9. Scheme [45] is 2350 bits, scheme [47] is 2980 bits, and scheme [27] is 3080 bits, while LMA touches the line for 2810 bits over several experiments. The first scheme has a low communication cost, while the other two have a higher communication cost. Calculating the average, it is 2803 bits, just near LMA.

C. Throughput Comparison

It is the frame in which a system completes the tasks or successfully delivers the data. We have also tested LMA with the other three schemes [45], [47], and [49] shown in Figure 11, and mapped the corresponding values in a graph, in which time in seconds is marked at the x-axis and throughput is on the y-axis in Figure 10. The LMA represents the maximum throughput compared to the other three schemes, indicating the system's scalability. With the increasing load inside the LMA, its processing is adapted to the load. Other schemes have their own implementation and level of throughput, but LMA is presenting better results in this aspect. From the average values of the throughput of these schemes, the value is 0.4866, while the LMA is 0.724, which is a 33.33% increase in this aspect.

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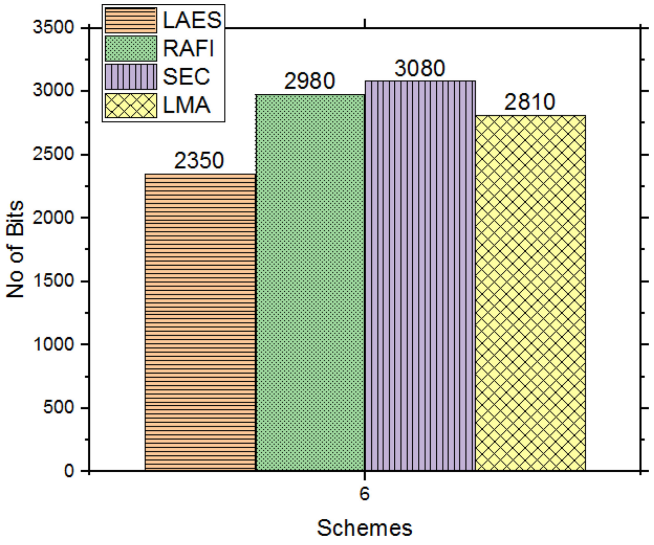


Fig. 10. LMA Communication Cost comparison with other Scheme.

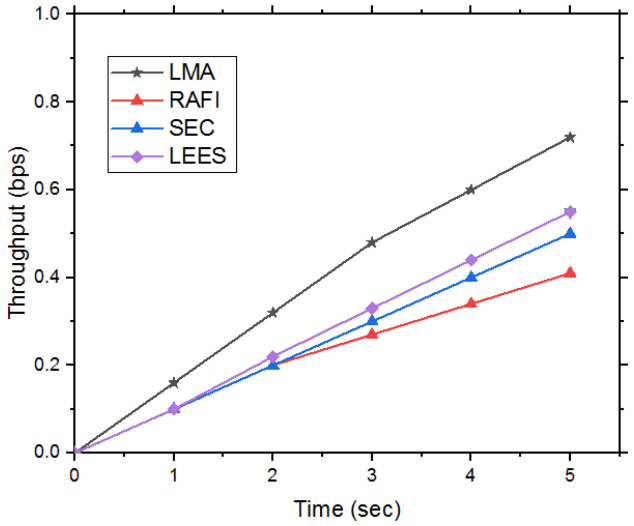


Fig. 11. LMA Throughput comparison with Other Schemes.

VII. Cryptanalysis of LMA

LMA was also checked by analyzing its behavior against different types of attacks and its resistance against malicious intruders.

A. Resists Against Man in the Middle Attack

The proposed LMA scheme incorporates cryptographic techniques, such as time stamps and hashing, to protect against man-in-the-middle attacks and maintain the confidentiality of sensitive information, especially in electronic data. In the communication process, the encrypted process of $En(xx)$ a malicious entity can intercept. However, due to the one-way nature of hashing, the actual value of ID(i) cannot be obtained. This helps prevent unauthorized entities from impersonating legitimate devices. During service usage, when a malicious device attempts to send a request while pretending to be another device, it sends the hashed values $ID(i)$ and $RP(xx)$. These values are obtained by masking the hashed value of $H(xx)$ with a timestamp. The masking process makes it challenging to uncover the original values as it relies on the intractability of the LMA scheme. As a result, the intruder cannot determine the values of $ID(i)$ or $H(xx)$. Therefore, the proposed scheme effectively thwarts attacks by insiders and ensures the system's integrity. The authentication process is shown in Algorithm 2.

Algorithm 2: Authentication Algorithm

Input: Start of the authentication process
Output: Detection of unauthorized access
Data: Communication Process, Interception of En(xx), Hashing (H(xx)), Masking with Time Stamp, Masked Hashed Values (ID(i) and RP(xx))
Result: Identification of unauthorized access

```
Start Communication Process Interception of En(xx)
 Hashing (H(xx)) Masking with Time Stamp Masked
 Hashed Values (ID(i) and RP(xx)) while Unauthorized
 Access? do
    if Yes then
        Intruder Determination of ID(i) and H(xx)
    else
        Continue Communication Process
    end
end
```

B. Resists Against Unauthorized Access

In LMA, integrating new and updated certificates enhances the security and authorization processes within the IoT network. The LMA algorithm, incorporating the issuance of certificates, ensures that only devices possessing valid and up-to-date certificates can access the services provided. Within the LMA framework, each device in the network is assigned a unique identifier. The server, denoted as *Svr*, generates a super secret key, *SVRkey*, while each device, *Sn*, is assigned a unique identifier, ID_{sn}. Each gateway GW_{sn} is assigned a unique identifier ID_{gw}. During the authentication process, devices present certificates verified by the server. Only devices with valid and up-to-date certificates are considered legitimate and granted access to the services. LMA will be more robust with the inclusion of this technique, and it will prevent unauthorized access where digital certificates are not in use, like the login process. This process has been shown in Algorithm 3 with distinct steps.

VIII. Conclusion

For connected electronic devices, the IoT provides the basis for offering customized services according to end-user preferences. IoT-based connected devices for e-commerce offer easy-to-use and reliable services, but the security of communication and the privacy of customers are the most challenging. Because of the pervasiveness and omnipresence of these electronic services, anyone can use them, easily compromising the authenticity of individual entities. On the other hand, normal authentication mechanisms are not feasible

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Algorithm 3: Authentication Algorithm for Resisting Unauthorized Access

Input: Start of the authentication process
Output: Detection of unauthorized access
Data: Integration of new and updated certificates, Unique identifiers for devices and gateways, Super secret key *SVRkey*

```
Generate super secret key SVRkey Assign unique
  identifiers ID_sn to each device Sn Assign unique
  identifiers ID_gw to each gateway GW_sn
while Authentication Process do
    foreach Device Sn do
        Obtain a certificate from server Svr if Certificate
          is valid and up-to-date then
            Grant access to services
        end
    end
end
```

due to the use of a wide array of devices and hybrid network structures. To address these challenges of privacy and authentication, a novel authentication solution called the LMA Scheme for Connected Devices in the IoT is devised". LMA uses a lightweight and automatic authentication procedure. Server-based architecture is deployed with a concurrent distributed approach. Although there is a slight increase in delay, LMA presents automatic and efficient authentication for all entities with low error rates. The proposed scheme addressed privacy and authentication concerns in IoT-based connected devices. The experimental findings are compared with the potential counterparts, and it is shown that the LMA decreases computation cost by about 56%, increases throughput by about 33.33%, and communication cost remains the same as the average of the three schemes. In evaluation, the results demonstrate that the presence of the LMA leads to a 20-millisecond increase in delay and a 2% decrease in PDR for 100 devices. Our findings contribute to the advancement of secure IoT authentication, enhancing IoT systems' overall security and reliability. However, while the accessibility of IoT services is a significant advantage, it also raises security and privacy concerns. Since anyone can potentially access and utilize these services, it becomes crucial to implement robust security measures to protect sensitive data, ensure secure authentication and authorization, and prevent unauthorized access or malicious activities. Conventional authentication mechanisms are impractical because of the heterogeneous devices, the network's amorphous structure, and constraints in computational and storage capacities. Furthermore, LMA's performance regarding authentication error rate is tested, and it is confirmed that its integration into the authentication system presents a reliable and efficient approach. It creates a low authentication error rate and improves system performance for connected electronic devices.

In the future, based on the findings of this study, we want to evaluate the scalability of the authentication scheme to accommodate a growing number of connected devices in the IoT ecosystem. We also need to investigate how the system performs under increasing load, considering different factors such as network traffic, Lidar Point Cloud Compression (L-PCC) technology, autonomous vehicles, device diversity, and dynamic changes in the IoT environment. Based on these findings, we want to propose an optimized approach to ensure efficient scalability and conduct a thorough security analysis to identify potential vulnerabilities and attack vectors in the proposed lightweight mutual authentication scheme. Moreover, to extend the authentication scheme's compatibility with a broader range of electronic platforms and standards, we also want to investigate how the system can seamlessly integrate with different protocols and frameworks commonly used in the electronic industry.

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